Day-26 Dec-28

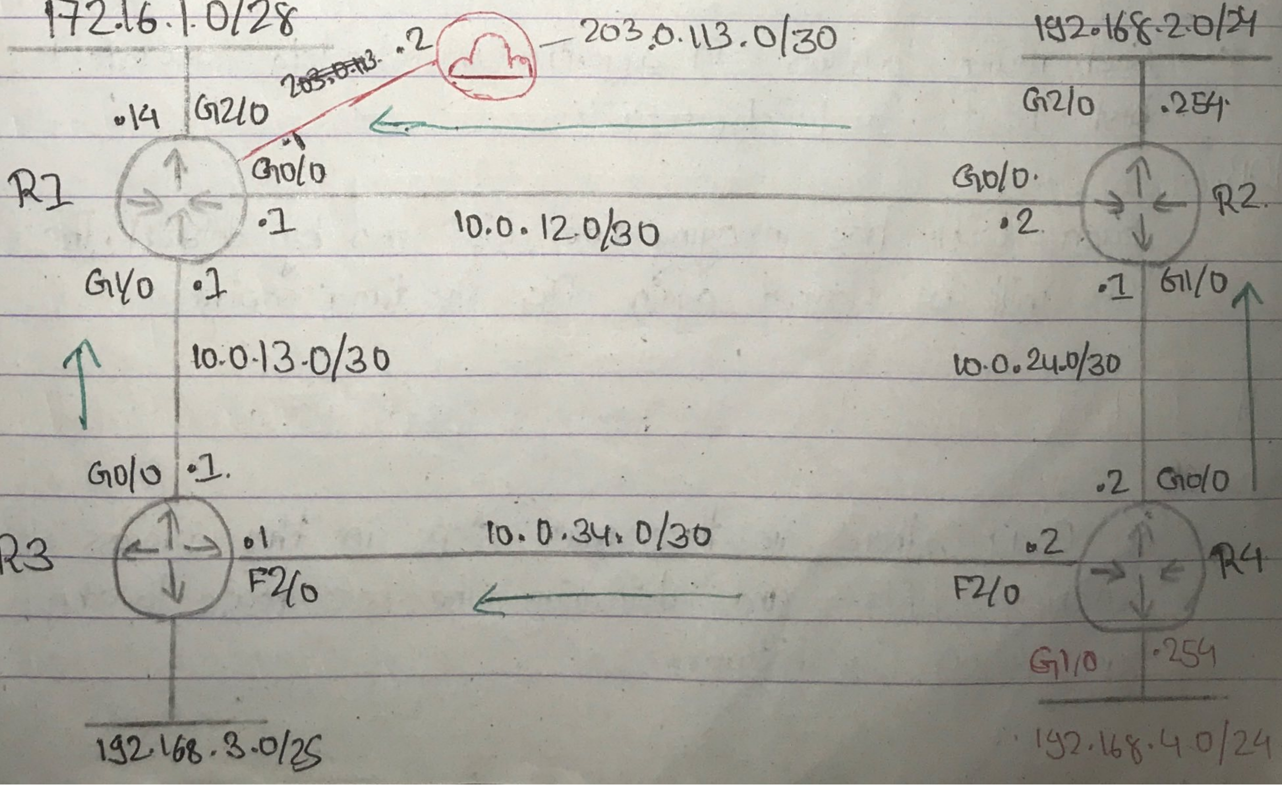
OSPF Part-I

OSPF

- Stands for Open Shortest Path First.
- Uses the Shortest Path First algorithm of Dutch computer scientist Edsger Dijkstra. (aka Dijkstra's algorithm ← remember this name)
- Three versions:
  - OSPFv1 (1989): OLD, not in use anymore
  - OSPF v2 (1998): Used for IPv4.
  - OSPF v3 (2008): Used for IPv6 (can also be used for IPv4, but usually v2 is used)

{CCNA मा OSPF v2 sodhxa as per version CCNA 1.1}

- Routers store information about the network in LSAs (Link State Advertisements), which are organized in a structure called the LSDB (Link State Database)
- Routers will flood LSAs untill all routers in the OSPF area develop the same map of the network (LSDB).



{Fig-I} {Network-I}

- OSPF is enabled on R4's G1/0 interface.
- R4 creates an LSA to tell its neighbors about the network on G1/0.

LSA.
RID - 4.4.4.4
IP: 192.168.4.0/24
cost: 1

R4 को कुनै पनि Interface मा 4.4.4.4 Configured छैन। So,
R4 ~~was~~ shall either have loopback of 4.4.4.4 or it was manually configured

- The network on the G1/0 interface is, of course, included in the LSA, since that's the main purpose of LSA. R4's cost is also included

  {We'll learn how to calculate OSPF cost later}

4 - The LSA is then flooded throughout the network untill all routers have received it.

- Now OSPF is activated on R4's G1/0 interface, the new LSA is added to the LSDB. {LSDB is identical for all routers in OSPF area}

- Each router then uses SPF algorithm (Djikstra) to calculate its best route to 192.168.4.0/24.

NOB:

Each LSA has an aging timer (30 min by default). The LSA will be flooded again after the timer expires.

Basic Process of OSPF.

In OSPF, there are three main steps in the process of sharing LSAs and determing the best route to each destination in network.

SPF {Shortest Path First}

1) Become neighbors with other routers connected to same segment.
2) Each LSAs with neighbor routers.
3) Calculate the best routes to each destination, and insert them into the routing table.

OSPF Areas.

- OSPF uses areas to divide up the network.
- Small networks can be single-area without any negative effects on performance.

  For eg. the network we had in earlier page with 4 routers is a small network, When configuring OSPF in a network like this we can ~~g~~ use just a single OSPF 'area' and there won't be any ~~degradation~~ degradation of network performance.

- In a larger networks, a single-area design can have negative effects {For eg. in OSPF network with 500 routers with over 1000 ~~router~~ subnets. instead of 4 routers & few subnets, using just a single OSPF area would be bad idea}.

  Some -ve effects of using single-area design in larger network?
  - the SPF algorithm takes more time to calculate routes.
  - the SPF algorithm requires exponentially more processing power on routers.
  - the larger LSDB takes up more memory on routers.
  - any small change in network causes every router to flood LSAs and run the SPF algorithm again.

- By dividing a large OSPF network into several small areas, we can avoid the above -ve effects.

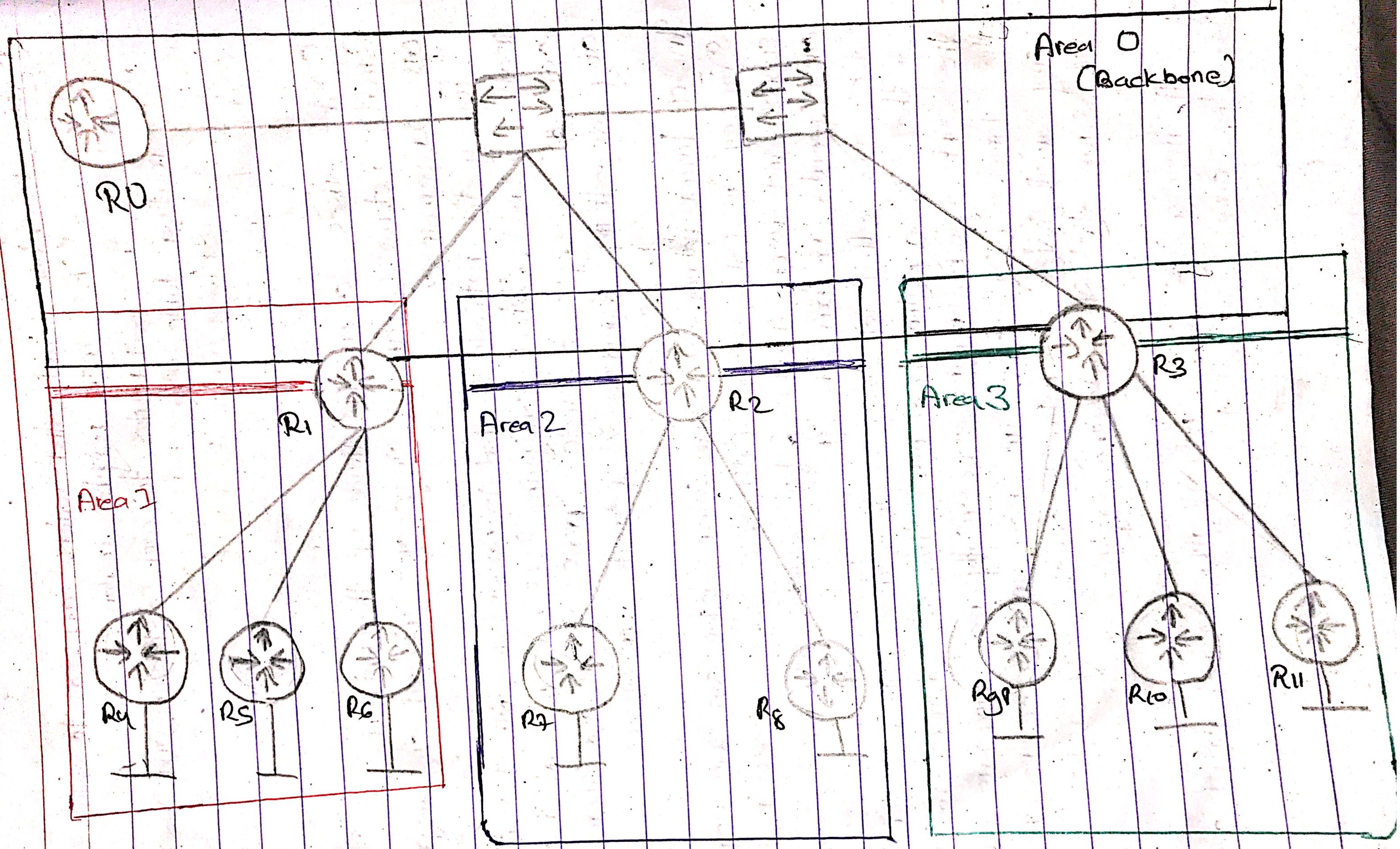

Area 0
(Backbone)
R0
R1
R2
R3
Area 1
Area 2
Area 3
R4
R5
R6
R7
R8
R9
R10
R11

Fig - II

- An area is a set of routers and links that share the LSDB.
- The ~~area~~ backbone area (area 0) is an area that all other areas must connect to.
  {For eg. Area 1 को Router (R1) चाहि Area 1 & 2 मा जोड्यो instead of 0 & 1 then. that OSPF design wouldn't be valid}

- Routers with all interfaces in the same area are called internal routers. In our e.g R0, R4, R5 ... etc are internal routers

- Routers with interfaces on multiple area are Area Border routers (ABRs).
  {In our eg R1, R2, R3}

NOB

ABRs maintain a separate LSDB for each area they are connected to. It is recommend that ~~you~~ we connect an ABR to a maximum of 2 areas. Connecting an ABR to 3+ areas can overburden the router.

- Routers connected to backbone area (area 0) are called backbone routers. It includes ABRs.. In our case R_0, R_1, R_2 & R_3 are backbone routers
- An intra-area route is a route to a destination inside the same OSPF area. {Eg: src in Area 1 to dst in area 1}. Eg. R1 - R4.
- An inter area route is a route to destination in a different OSPF area.. R_6 to R_7.

Rules about OSPF areas

- OSPF areas should be contiguous.

  Eg. If in our network earlier if there was Area 1 even in place of Area 3, then ~~&~~ do Area 1 won't be contiguous. as half-area is one-place & half on other

- All OSPF areas must have at least one ABR connected to backbone area.
- OSPF interfaces in the same subnet must be in same area. As if they're not in same, they won't become OSPF ~~a~~neighbor & won't exchange info about networks they know about.

Basic OSPF Configuration

We'll use Network in Figure - I. All ~~a~~ routers are in OSPF Area 0.

Let's configure on R1.

```
R1 (config)# router ospf
    <1-65535> Process ID.
 R1 (config) # router ospf 1
 R1(config-router)# network 10.0.12.0 0.0.0.3 area 0.
  R1(config-router) # ~~10.0.13.0~~ network 10.0.13.0 0.0.0.3 area 0
R1 (config-router) # network 172.16.1.0 0.0.0.15 area 0
```

- A router can run multiple OSPF process at once, and this (process id) ID is used in the router to identify each of them. We'll typically use single OSPF process

For EIGRP routers to become neighbors, their AS number has to match. However OSPF proces ID is different. OSPF process id is locally significant (within that router only).

- Routers with different process IDs can become OSPF neighbors
- The OSPF network command requires us to specify the area.

For CCNA, we basically need to configure single-area OSPF (area 0).

NOB repeated

The network command is simply used to tell the router which interfaces to activate OSPF on. It doesn't tell the routers advertise these networks.

The network command tells OSPF to.

- look for any interfaces with an IP address contained in the range specified in the network command.
- Activate OSPF on the interface in the specified area.
- The router will then try to become OSPF neighbors with the OSPF-activated neighbor routers.

Passive interface command.

```
R1(config-router)# passive-interface g2/0
```

{It's totally same as in EIGRP}

The passive-interface command tells the router to stop sending OSPF 'hello' messages out of interface.

- However the router will continue to send LSAs informing its neighbors about the subnet configured on interface.

- We should always this command on interfaces which don't have any OSPF neighbors

Advertise a default route into OSPF.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2
R1(config-router)# default-information originate.
```

show ip protocols {OSPF perspective}

Router ID order priority:
1) Manual configuration.
2) Highest IP address on a loopback interface.
3) Highest IP address on a physical interface.

Manually configuring router-id.

```
R1(config-router)# router-id 1.1.1.1.
```

{In EIGRP it's eigrp router-id but in OSPF its just router-id}

To make 1.1.1.1 on effect we either need to reload router or use the command as below.

```
R1# clear ip ospf process
```

{bad idea as for short time R1 looses all routes & won't forward to destination. Use in lab but in real its bad idea]

ASBR (Autonomous System Boundary Router)

It is an ~~ASBR~~ OSPF router that connects the OSPF network to external network.

R1 is connected to internet. By using default-information ~~org~~ originate command, R1 becomes an ASBR.

Unlike EIGRP, OSPF doesn't support unequal-cost load-balancing, but it does support ECMP load-balancing over a maximum of 4 paths by default.

To change Maximum paths.

```
R1(config-router)# maximum-paths 8
```

distance (default 110) (Administrative ~~Co~~ Distance)

To chang use same cmmnd as RIP & EIGRP

~~R1(config-router)# maximum-paths~~ ?

~~R1(config-router)~~

```
R1(config-router)#distance 85
```